

INDIGA, Russian Federation

WMO: 222920

Lat: 67.689N

Lon: 48.684E

Elev: 4

StdP: 101.27

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-29.7	-35.2	0.1	-32.1	-32.6	0.2	-29.7	20.8	-10.7	18.5	-7.5	2.7	110	0.970

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	21.6	16.6	19.4	15.4	17.5	14.6	17.3	20.7	16.2	18.8	14.8	16.9	5.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.4	19.2	14.8	10.5	17.7	13.5	9.6	16.5	49.0	20.6	45.5	18.9	41.5	16.9	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.0	14.2	12.8	DB	-34.8	25.2	2.3	2.1	-36.5	26.7	-37.8	27.9	-39.1	29.1	-40.8	30.6	(4)
			WB	-34.9	18.9	2.3	1.7	-36.5	20.1	-37.9	21.1	-39.2	22.0	-40.9	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-13.7	-11.4	-9.6	-4.2	0.5	6.6	11.6	10.7	7.0	1.1	-6.3	-10.0
	DBStd	10.36	7.94	8.24	7.10	4.83	4.56	4.35	4.53	3.22	3.07	3.96	6.49	7.67
	HDD10.0	4332	735	600	608	426	296	122	36	28	97	275	490	621
	HDD18.3	7216	993	833	866	676	552	352	214	237	339	533	740	879
	CDD10.0	164	0	0	0	0	2	19	84	51	8	0	0	0
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	23	0	0	0	0	0	2	19	2	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	6.4	7.0	7.0	6.8	6.4	5.7	5.6	5.3	5.8	6.0	7.0	7.2	7.1
Precipitation	PrecAvg	416	32	25	20	19	21	42	44	52	52	55	29	24
	PrecMax	705	101	75	44	35	75	132	135	116	108	130	48	58
	PrecMin	219	8	3	2	3	5	6	2	11	18	17	15	7
	PrecStd	101	22	15	11	9	13	29	29	27	25	25	9	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.2	1.5	7.1	17.7	22.2	25.8	22.6	16.5	9.5	2.8	1.4
		MCWB	-0.3	0.8	0.6	4.3	12.0	16.3	18.0	17.9	13.8	8.3	2.0	0.7
	2%	DB	-0.9	0.4	0.8	3.6	13.3	18.9	22.9	19.7	14.4	7.9	2.1	0.7
		MCWB	-1.4	0.1	0.2	1.9	9.7	14.0	17.3	16.0	12.2	7.2	1.5	0.2
	5%	DB	-2.1	-0.6	0.0	2.3	9.6	16.2	20.6	17.8	12.8	6.9	1.3	0.0
		MCWB	-2.6	-1.0	-0.5	1.1	7.0	12.6	16.3	15.1	11.1	6.1	0.8	-0.4
	10%	DB	-3.3	-1.7	-1.0	1.4	6.7	13.6	18.6	15.9	11.3	5.9	0.7	-1.0
		MCWB	-3.8	-2.2	-1.5	0.6	4.9	11.1	15.3	14.0	10.1	5.0	0.2	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.9	0.9	4.6	12.5	17.1	19.2	18.8	14.5	8.6	2.4	1.0
		MCDB	0.1	1.0	1.2	7.1	16.6	20.8	23.2	22.3	16.0	9.1	2.6	1.1
	2%	WB	-1.3	0.1	0.3	2.0	9.6	14.4	17.9	16.7	12.7	7.2	1.6	0.4
		MCDB	-0.9	0.3	0.7	3.0	13.3	17.4	21.8	18.9	14.0	7.8	1.8	0.6
	5%	WB	-2.5	-0.9	-0.4	1.3	7.1	12.7	16.9	15.4	11.2	6.1	0.9	-0.4
		MCDB	-2.1	-0.7	0.0	2.0	9.4	15.9	20.1	17.3	12.3	6.8	1.2	-0.1
	10%	WB	-3.7	-2.1	-1.4	0.7	4.9	11.0	15.7	14.1	10.2	5.1	0.3	-1.4
		MCDB	-3.3	-1.7	-1.0	1.3	6.7	13.8	18.1	15.8	11.2	5.8	0.7	-1.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	5.9	6.1	5.5	5.1	5.7	5.6	5.1	4.1	3.4	4.9	5.8
		MCDBR	6.2	4.1	3.3	5.5	10.8	11.2	9.8	8.8	6.6	3.7	3.8	3.6
		MCWBR	6.3	4.1	3.1	4.5	7.2	6.5	4.9	5.7	4.9	3.6	3.5	3.8
	5% WB	MCDBR	6.3	4.4	3.2	4.9	10.7	10.2	9.0	7.9	5.8	3.6	3.6	3.6
		MCWBR	6.3	4.4	3.2	4.2	7.3	6.9	5.0	5.7	4.4	3.5	3.6	3.8
Clear-Sky Solar Irradiance	taub	0.213	0.258	0.299	0.329	0.339	0.341	0.375	0.353	0.324	0.288	0.366	N/A	
	taud	1.903	2.030	2.087	2.147	2.331	2.477	2.413	2.459	2.491	2.248	2.581	N/A	
	Ebn at Noon	98	529	723	812	853	864	814	794	728	505	121	0	
	Edh at Noon	11	52	88	110	103	92	96	82	62	42	13	0	
All-Sky Solar Radiation	RadAvg	0.04	0.49	1.96	3.76	4.84	5.08	4.82	3.17	1.64	0.51	0.08	0.00	
	RadStd	0.00	0.06	0.20	0.29	0.28	0.61	0.57	0.32	0.18	0.08	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47)	Regional (3 neighbors)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page